Lecture 4 // Sept 3 // ASD connections & Gauge

Recall

To prove Donaldson's theorem, we need to study the space of anti-self-dual connections on M modulo gauge. Last time, we defined connections and and discussed the codifferential and the Hodge-Laplacian, and stated the main theorem of Hodge theory.

- $d^* = (-1)^k \star d \star$
- $\Delta := (d + d^*)^2 = dd^* + d^*d$
- $\Omega^k = \operatorname{im} d \oplus \operatorname{im} d^* \oplus \mathcal{H}^k$ where $\mathcal{H}^k := \ker d \cap \ker d^*$.

Today, we'll discuss self duality.

First, note that the statement of the Hodge decomposition given above isn't quite the same as from last time. It's not too difficult to verify that the decomposition $\Omega^k = \operatorname{im} d \oplus \operatorname{im} d^* \oplus \mathcal{H}^k$ are orthogonal, ie, have pairwise intersection $\{0\}$. However, it is harder to show that $\Omega^k = \operatorname{span}\{\mathcal{H}^k, \operatorname{im} d, \operatorname{im} d^*\}$. For this, you'll need some functional analysis, so we'll only verify the first part.

- If $\alpha \in \mathcal{H}^k \cap \operatorname{im} d$, $\Delta\alpha = 0$, so $\langle \alpha, \alpha \rangle = \langle \alpha, d\beta \rangle = \langle d^*\alpha, \beta \rangle = 0$.
- if $\alpha \in \operatorname{im} d$ and $\alpha \in \operatorname{im} d^*$, so $\langle \alpha, \alpha \rangle = \langle d\beta, d^*\gamma \rangle = \langle d^2\beta, \gamma \rangle = 0$.

Each cohomology class is in the kernel of d, and since $\ker(d) \cong \operatorname{im}(d) \oplus \mathcal{H}^k$, each class is in $\operatorname{im}(d) \oplus \mathcal{H}^k$. Therefore there can be only one harmonic representative. In particular, the different cohomology classes are affine cosets of $\operatorname{im}(d)$.

We'll now specialize to (M^4, g) and let $k = 2$. Since $n - k = 2$, the Hodge star gives an involution $\star : \Omega^2 \to \Omega^2$ which squares to the identity: $\star^2 = (-1)^{2(4-2)} = +1$. This means we can look at the positive and negative eigenspaces of $\star$ on Ω^2.

Definition 1

If $\star\alpha = \alpha$, α is called self-dual (SD). If $\star\alpha = -\alpha$, α is anti-self-dual (ASD). The subspace of self-dual 2-forms is denoted Ω^2_+, and the subspace of anti-self-dual 2-forms is Ω^2_-.

After choosing an orthonormal basis for $T^*_pM = \operatorname{span}\{e_1^*, e_2^*, e_3^*, e_4^*\}$, we get that

$$\Omega^2 = \operatorname{span}\{e_1 \wedge e_2, e_1 \wedge e_3, e_1 \wedge e_4, e_2 \wedge e_3, e_2 \wedge e_4, e_3 \wedge e_4\}.$$

To find the eigenspaces of $\star$, we'll compute $\star(e_i \wedge e_j$ for each basis element:

$$\star(e_1 \wedge e_2) = e_3 \wedge e_4, \qquad \star(e_1 \wedge e_3) = -e_2 \wedge e_4, \qquad \star(e_1 \wedge e_4) = e_2 \wedge e_3,$$

$$\star(e_2 \wedge e_3) = e_1 \wedge e_4, \qquad \star(e_2 \wedge e_4) = -e_1 \wedge e_3, \qquad \star(e_3 \wedge e_4) = e_1 \wedge e_2.$$

Therefore

$$\Omega^2_+ = \operatorname{span}\{e_1 \wedge e_2 + e_3 \wedge e_4,\ e_1 \wedge e_3 - e_2 \wedge e_4,\ e_1 \wedge e_4 + e_2 \wedge e_3\}$$

and

$$\Omega^2_- = \text{span}\{e_1 \wedge e_2 - e_3 \wedge e_4,\ e_1 \wedge e_3 + e_2 \wedge e_4,\ e_1 \wedge e_4 - e_2 \wedge e_3\},$$

which splits the space of 2-forms as $\Omega^2 = \Omega^2_+ \oplus \Omega^2_-$

Further, note that if α is self-dual, $\int_M \alpha \wedge \alpha = \int_M \alpha \wedge \star\alpha = \langle \alpha, \alpha \rangle = \|\alpha\|^2$. Similarly, if α is anti-self-dual, $\int_M \alpha \wedge \alpha = \int_M \alpha \wedge - \star \alpha = -\int_M \alpha \wedge \star\alpha = -\langle \alpha, \alpha \rangle = -\|\alpha\|^2$, so self-dual 2-forms have positive self cup product, and anti-self-dual 2-forms have negative self cup product.

Classification of Bundles

Previously, we discussed vector bundless and connections on vector bundles. The bundles $E \xrightarrow{p} B$ had transition functions $g \in \text{GL}_n(\mathbb{C})$, and connections $\nabla = d + A$ were defined locally as matricies with entries given by 1-forms. Now, we'll specify to SU(2)-connections.

Definition 2 (G-vector bundle)
Let G be a Lie group; think of it very explicitly as a Lie subgroup of $\text{GL}_n(\mathbb{C})$. Then the Lie algebra $\mathfrak{g}$ is a subspace of $\text{Mat}_n(\mathbb{C})$. We say that a vector bundle has structure group G if the transition functions $g \in \text{GL}_n(\mathbb{C})$ are valued in the subgroup $G \subset \text{GL}_n(\mathbb{C})$: $g \in G \subseteq \text{GL}_n(\mathbb{C})$-valued. A vector bundle with this property is called a G-vector bundle.

Definition 3 (G-connection)
Given a G-vector bundle, a G-connection on E is a connection whose associated connection 1-form A is valued in $T^*M \otimes \mathfrak{g}$. This means that A is again a matrix of 1-forms, but the matrix must lie in Lie algebra. Even more concretely, given any tangent vector, we may feed it to A. This results in a matrix of numbers–a literal element of $\text{Mat}_n(\mathbb{C})$–which is required to be in $\mathfrak{g} \subseteq \text{Mat}_n(\mathbb{C})$).

Exercise 4
Check that the definition of a G-connection is preserved by coordinate change

$$A \mapsto g^{-1}Ag + g^{-1}dg \quad \text{(as long as } g \in G\text{)}$$

e.g., if $A \in \mathfrak{g}$, $g^{-1}Ag + g^{-1}dg \in \mathfrak{g}$:

$\hookrightarrow$ $\mathfrak{g}$ closed under conjugation with $g \in G$

$\hookrightarrow$ dg in $T_eG \rightsquigarrow$ must move to e before evaluating, so precompose with g^{-1}

Exercise 5
If $\nabla = d + A$ is a G-connection, $F_A = dA + A \wedge A \in \Lambda^2 T^*M \otimes \mathfrak{g}$ at each point, i.e., for any v, w, feed in v, w to 2-form entries in matrix, get a matrix in $\mathfrak{g} \in \text{Mat}_n(\mathbb{C})$.

F_A can equivalently be written as

$$F_A \in \Gamma\big(\Lambda^2 T^*M \otimes \underbrace{\mathrm{ad}(E)}_{\text{adjoint bundle}}\big)$$

Before, $F_A \in \Gamma(\Lambda^2 T^*M \otimes \mathrm{End}(E))$. $\mathrm{ad}(E) \subset \mathrm{End}(E)$ is the subbundle of $\mathrm{End}(E)$ with all transition functions in $\mathfrak{g}$; $\mathfrak{g}$ is closed under conjugation, so transition functions on $\mathrm{End}(E)$ restrict to those on $\mathrm{ad}(E)$.

We write $\Omega^2(\mathrm{ad}\, E) := \Gamma(\Lambda^2 T^*M \otimes \mathrm{ad}(E))$.

In our setting, there will be 2 G's which are relevant:

① $G = \mathrm{U}(1) =$ complex line bundles $\subset \mathrm{GL}_1(\mathbb{C})$

so g is S^1 valued

so tangent space at 1 in U(1) is imaginary axis $\Rightarrow A \in i\,\Omega^1(M;\mathbb{R})$

② $G = \mathrm{SU}(2) \subseteq \mathrm{GL}_2(\mathbb{C})$

so g looks like $\begin{pmatrix} a & -\bar{b} \\ b & \bar{a} \end{pmatrix}$, $|a|^2 + |b|^2 = 1$

so A is locally in $\begin{pmatrix} \alpha & -\bar{\beta} \\ \beta & -\alpha \end{pmatrix}$, α is purely imaginary 1-form

$\mathfrak{su}(2) =$ trace-free skew-hermitian matrices.

How to classify these G-bundles?

Recall Chern–Weil: For U(1) bundles, only 1 chern class c_1.

Theorem 6
$c_1 \in H^2(M;\mathbb{Z})$ classifies U(1) bundles $/M$.

For SU(2), have c_1 and c_2. If F_A is 2×2, check that:

$$c_1 = \frac{-1}{2\pi i}\,\mathrm{tr}(F_A) \quad (\text{coeff. of } t) \;\rightsquigarrow\; \text{in } \det\big(I - \tfrac{t}{2\pi i} F_A\big)$$

$$c_2 = \frac{-1}{4\pi^2}\det(F_A) \quad (\text{coeff. of } t^2)$$

Specifically, for an SU(2) connection, we know that

$$c_1 = \frac{-1}{2\pi i}(\alpha + -\alpha) = 0$$

$$c_2 = \frac{-1}{4\pi^2}\left(\alpha \wedge -\alpha - \beta \wedge -\bar{\beta}\right)$$

For an SU(2) bundle, also have that

$$c_2 = \frac{1}{8\pi^2}\,\mathrm{Tr}(F_A \wedge F_A)$$

$$\left(F_A \wedge F_A = \begin{pmatrix} \alpha & -\bar{\beta} \\ \beta & -\alpha \end{pmatrix} \wedge \begin{pmatrix} \alpha & -\bar{\beta} \\ \beta & -\alpha \end{pmatrix} = \begin{pmatrix} \alpha \wedge \alpha - \bar{\beta} \wedge \beta & * \\ * & -\beta \wedge \bar{\beta} + \alpha \wedge \alpha \end{pmatrix}\right)$$

Theorem 7
SU(2) bundles over M^4 are classified by c_2.
Not true in general: $\pi_k(B\,\mathrm{SU}(2)) \neq 1 \;\forall\; k > 4$.

In particular, a (closed oriented) 4-manifold has $\mathbb{Z}$-many SU(2) bundles, since $c_2 \in H^4(M;\mathbb{Z}) \cong \mathbb{Z}$.

Definition 8
Given an SU(2) bundle, the "instanton number" = "2nd chern number" of E is

$$k = -\int_M c_2 \quad (= -\langle [M], c_2\rangle, \text{ convention})$$

Splittings of Bundles

Let E be an SU(2) bundle. When does E split as $E = L_1 \oplus L_2$?, $L_i = \mathrm{U}(1)$ bundle, transition functions h split as $h = \begin{pmatrix} h_{11} & 0 \\ 0 & h_{22} \end{pmatrix}$.

If E splits, it's called a split bundle, $L_1 \oplus L_2$ is a "splitting." If E is split, a split connection is locally of the form $A = \begin{pmatrix} \alpha_1 & 0 \\ 0 & \alpha_2 \end{pmatrix}$.

If E is an SU(2) bundle, any splitting into U(1) bundles must be of the form $L \oplus \bar{L}$, since $\alpha_1 = -\alpha_2$

$$\Rightarrow \text{a split SU(2) connection looks like } A = \begin{pmatrix} \alpha & 0 \\ 0 & -\alpha \end{pmatrix}$$

Question 9. *If E splits, how many splittings?*

If E splits, $F_A = \begin{pmatrix} d\alpha & 0 \\ 0 & -d\alpha \end{pmatrix}$ since $A \wedge A = 0$.

$$\Rightarrow \text{if } E \text{ splits as } L \oplus \bar{L}, \quad c_2(E) = c_1(L) \cup c_1(\bar{L}) = c_1(L) \cup -c_1(L)$$

$$\Rightarrow \text{if } E \text{ splits, determined by 1 line bundle, which is determined by one } H^2 \text{ class}$$

$$\Rightarrow k = -\langle c_2, [M]\rangle = -\langle c_1 \cup -c_1, [M]\rangle = \langle c_1 \cup c_1, [M]\rangle$$

Lemma 10
Let E be an SU(2) bundle with instanton # k over M^4. Then E splits iff there exists $\alpha \in H^2(M,\mathbb{Z})$ such that $\int_M \alpha^2 = k$, and the # splittings = $\frac{1}{2}$# such α (if H^2 torsion free).

The Punchline:

Definition 11
Let (M^4, g) be a Riemannian 4-manifold and E be an SU(2) bundle over M (will usually consider $k = 1$).

$$\mathcal{A} := \{\text{SU(2) connections on } E\}$$

this is an ∞-dimensional affine space over $\Gamma(T^*M \otimes \operatorname{ad} E)$.

$$\mathcal{G} := \{\text{SU(2) gauge group}\} = \Gamma(\operatorname{Aut}(E))$$

section in $\operatorname{End}(E)$ valued in $\operatorname{ad}(E)$ (locally).

The gauge group acts on the space of connections: $\mathcal{G} \curvearrowright \mathcal{A}$ by

$$g \curvearrowright A = g^{-1}Ag + g^{-1}dg$$

$\mathcal{G}$ is similar to $[M, \mathrm{SU}(2)]$. It's not quite; this is only true if E is trivial.

Definition 12
A connection $\nabla = d + A$ is self-dual if $F_A = \star F_A$, anti-self-dual if $F_A = -\star F_A$

(extend $\star$ to act on $\Gamma(\Lambda^2 T^*M \otimes \operatorname{ad}(E))$ by acting on just the 2-forms part; e.g., $\star$ of every entry of matrix.)

(Really, should say "connection with self-dual curvature.") Check: property of being self dual is preserved by gauge transformations: $F_A \mapsto g^{-1}F_A g$, $\star$ commutes with conjugation. E.g., the property of being SD is invariant under $\mathcal{G}$.

Definition 13
The (A)SD Moduli Space is

$$\mathcal{M} := \{\text{(anti)-self-dual SU(2) connections}\}/\mathcal{G} \subset \mathcal{A}/\mathcal{G}$$